LAB-8: Draw the Component Diagram

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Project: AI Chatbot for University Helpdesk (RAG-based) |
github.com/Xxierrr/AI-Chatbot-for-University-Helpdesk

1. Aim

To draw a UML component diagram of the University Helpdesk Chatbot that shows its major software components, their provided/required interfaces and dependencies.

2. Theory

A **component diagram** shows the static, modular structure of a system. A component is a replaceable, self-contained unit that exposes interfaces and hides its internals. It is drawn as a rectangle with two small tabs on its left edge, with an optional stereotype such as `<<database>>` or `<<external>>`. Arrows show a dependency: the component at the tail uses the one at the head.

3. Component Diagram

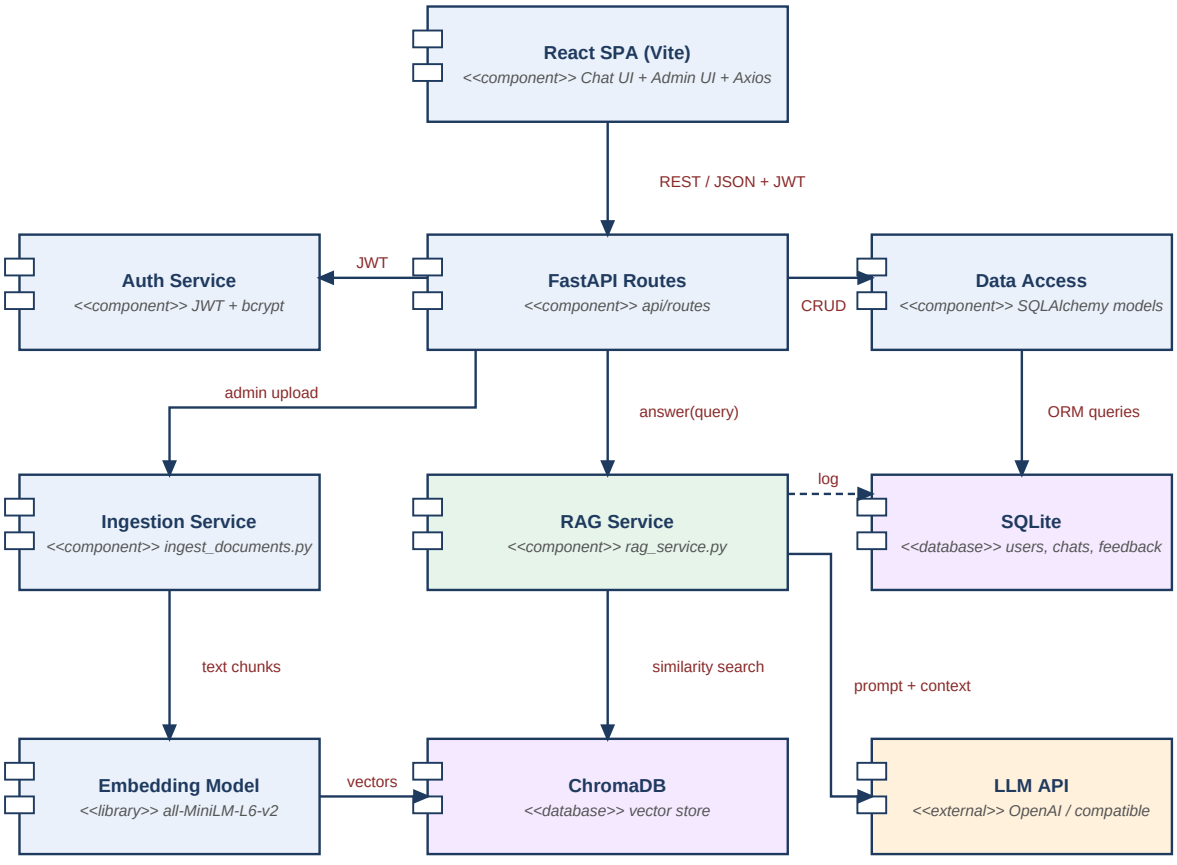

Figure 8.1 – Component diagram of the AI Chatbot for University Helpdesk (dashed arrow = secondary dependency)

4. Component Description

Component	Responsibility	Technology	Depends on
React SPA	Chat interface, suggested questions, feedback buttons, history and admin dashboard.	React 18, Vite, Tailwind, Axios	FastAPI Routes
FastAPI Routes	Exposes REST endpoints (/api/chat, /api/auth/*, /api/admin/*, /api/feedback, /api/health) and enforces role-based access.	FastAPI, Pydantic	Auth, Data Access, RAG, Ingestion
Auth Service	User registration, login, JWT creation/verification, password hashing.	JWT, bcrypt	Data Access
Data Access	ORM models for users, conversations, messages, feedback and documents.	SQLAlchemy	SQLite
RAG Service	Embeds query, retrieves top-k chunks, builds the prompt, calls the LLM and returns a cited answer.	LangChain	ChromaDB, Embedding Model, LLM API
Ingestion Service	Parses uploaded files, splits into chunks, and stores embeddings.	ingest_documents.py	Embedding Model, ChromaDB
Embedding Model	Converts text into 384-dimensional vectors.	sentence-transformers	–
ChromaDB / SQLite	Vector store for knowledge base / relational store for application data.	ChromaDB, SQLite	–
LLM API	External service that generates the natural-language answer.	OpenAI or compatible	–

5. Interfaces

- **Provided by FastAPI Routes:** REST/JSON interface consumed by the React SPA.
- **Provided by RAG Service:** an *answer(query, history)* interface used by the chat route.
- **Required by RAG Service:** vector-search interface (ChromaDB) and chat-completion interface (LLM API).

6. Conclusion

The component view shows a layered, loosely coupled design: the RAG logic is isolated in one service, so the LLM provider, the vector database or the UI can be replaced without rewriting the remaining components.